M3503: Lecture notes (Nchijane JN)

Differential eqns have many applications in physics, engineering, chemistry, biology, physiology and economics. Problems which arise in these areas involve mathematical modelling of physical phenomena using differential eqns.

Differential eqns (DEs) are broken down into ordinary differential eqns (ODEs) and partial differential eqns (PDEs).

If there is a single independent variable then the derivatives are ordinary and an eqn is called an ODE.

If there are two or more independent variables there the derivatives are partial and an eqn is a PDE.

Examples of DEs:

1) $\frac{dy}{dx} = \cos(x)$ (ODE)

2) $\frac{dy}{dx} = x + 1$ (ODE)

3) $L\frac{d^2i}{dt^2} + R\frac{di}{dt} + \frac{1}{C}i = E\omega\cos(\omega t)$ (ODE)

4) $\frac{d^2z}{dx^2} + \frac{d^2z}{dy^2} = x^2 + y$ (PDE)

5) $(x^2+y^2)\,dx - 2xy\,dy = 0$ (ODE)

6) $\frac{d^2y}{dt^2} + a\frac{dy}{dt} + ky = 0$ (ODE; a & k are parameters)

7) $\frac{\partial y}{\partial x} + \frac{\partial y}{\partial y} = x - 2y$, (PDE)

First order ODEs:

These are eqns of the form

$$F(x, y, y') = 0 \quad \text{or} \quad y' = f(x, y).$$

The eqn is linear if it's expressible in the form:

$$a_1(x)\,y' + a_0(x)\,y = g(x).$$

In this section, we will deal with the following types of 1st order ODEs:

(a) Variable separable eqns:

An eqn of the form $y' = f(x, y)$ is called separable if it's expressible in the form $y' = g(x)f(y)$.

Solving a variable separable eqn:

$$y' = g(x)f(y) \Rightarrow \frac{dy}{dx} = g(x)f(y)$$

$$\Rightarrow \frac{dy}{f(y)} = g(x)\,dx$$

$$\Rightarrow \int \frac{dy}{f(y)} = \int g(x)\,dx + C$$

Examples:

i) Solve $y' = \frac{x^2-1}{y^2}$, $y \neq 0$

Soln: $\frac{dy}{dx} = \frac{x^2-1}{y^2} \Rightarrow y^2 dy = (x^2-1)\,dx$

$$\Rightarrow \int y^2 dy = \int (x^2-1)dx$$

$$\Rightarrow \frac{y^3}{3} = \frac{x^3}{3} - x + C \Rightarrow y^3 - x^3 + 3x = C_1$$

ii) Solve $y \sin(x)\, e^{\cos(x)}\, dx + y^{-1} dy = 0$

Soln: $y \sin(x)\, e^{\cos(x)}\, dx + y^{-1} dy = 0$

$$\Rightarrow \sin(x)\, e^{\cos(x)}\, dx = -\frac{dy}{y^2}$$

$$\Rightarrow -e^{\cos(x)} \cdot d(\cos(x)) = \frac{1}{y} + c$$

$$\Rightarrow -e^{\cos(x)} - y^{-1} = C \Rightarrow y = \frac{1}{C - e^{\cos(x)}}$$

iii) Solve $y' = \frac{xy + 3x - y - 3}{xy - 2x + 4y - 8}$

Soln: $\frac{dy}{dx} = \frac{x(y+3) - 1(y+3)}{x(y-2) + 4(y-2)} = \frac{(x-1)(y+3)}{(x+4)(y-2)}$

$$\Rightarrow \frac{y-2}{y+3}\, dy = \frac{x-1}{x+4}\, dx$$

$$\Rightarrow \left(\frac{y}{y+3} - \frac{2}{y+3}\right) dy = \left(\frac{x}{x+4} - \frac{1}{x+4}\right) dx$$

$$\Rightarrow \left(1 - \frac{5}{y+3}\right) dy = \left(1 - \frac{5}{x+4}\right) dx$$

$$\Rightarrow y - 5 \ln|y+3| = x - 5 \ln|x+4| + c$$

$$y - x + 5\left(\ln|x+4| - \ln|y+3|\right) = c$$

$$\Rightarrow y - x + 5\left(\ln\left|\frac{x+4}{y+3}\right|\right) = c$$

Eqns of the form: $y' = g(ax+by) \cdots (2)$

Let $z = ax+by \Rightarrow z' = a + by'$

$$\Rightarrow y' = \frac{z'-a}{b}$$

eqn (2) becomes:

$$\frac{z'-a}{b} = g(z)$$

$$\Rightarrow z' = b\,g(z) + a$$

$\Rightarrow dz = (b\,g(z)+a)\,dx$ which is a variable separable eqn again.

(b) Eqns with Homogeneous coefficients.

The fn $f(x,y)$ is said to be homogeneous of degree k in x and y if:

$$f(\lambda x, \lambda y) = \lambda^k f(x,y)$$

e.g. Let $f(x,y) = x^2 + y^2$

$$\Rightarrow f(\lambda x, \lambda y) = (\lambda x)^2 + (\lambda y)^2 = \lambda^2 x^2 + \lambda^2 y^2$$

$$\Rightarrow f(\lambda x, \lambda y) = \lambda^2(x^2+y^2) = \lambda^2 f(x,y)$$

$\therefore f(x,y)$ is homogeneous of degree 2 in x and y.

Theorem 1:

If $M(x,y)$ and $N(x,y)$ are both homogeneous and of the same degree in x and y, then the fn $\frac{M(x,y)}{N(x,y)}$ is homogeneous of degree zero in x and y.

Proof:

Let $M(\lambda x, \lambda y) = \lambda^k M(x,y)$

and $N(\lambda x, \lambda y) = \lambda^k N(x,y)$

$$\therefore \quad f(\lambda x, \lambda y) = \frac{M(\lambda x, \lambda y)}{N(\lambda x, \lambda y)} = \frac{\lambda^k M(x,y)}{\lambda^k N(x,y)}$$

$$= \lambda^{k-k} \frac{M(x,y)}{N(x,y)}$$

$$= \lambda^0 \frac{M(x,y)}{N(x,y)}$$

$\therefore \frac{M(x,y)}{N(x,y)}$ is homogeneous of degree zero.

Theorem 2:

If $f(x,y)$ is homogeneous of degree zero in x and y, then $f(x,y)$ is a fn of $\frac{y}{x}$.

Proof: Let $y = vx$, then $f(x,y) = f(x, vx) = x^0 f(1,v)$
$= f(1,v) = f(v)$.

Solving eqns with homogeneous coefficients:

Let M and N such that

$$M(x,y)\,dx + N(x,y)\,dy = 0, \quad \text{where} \quad \ldots (3)$$

M and N are homogeneous of the same degree in x and y.

$\therefore$ eqn (3) can be written as

$$y' + \frac{M(x,y)}{N(x,y)} = 0$$

$$\Rightarrow y' + g\left(\frac{y}{x}\right) = 0 \quad \ldots (4)$$

Now let $y = vx \Rightarrow y' = v + v'x$

Then eqn (4) becomes

$$v + v'x + g(v) = 0$$

$$\Rightarrow v'x = -g(v) - v$$

$$\Rightarrow x\frac{dv}{dx} = -(g(v) + v)$$

$$\Rightarrow \frac{1}{(g(v)+v)}\,dv = \frac{dx}{x}, \text{ which}$$

is a variable separable eqn.

(c) Eqns with linear coefficients:

Consider the f.f. type of eqn:

$$(a_1x + b_1y + c_1)\,dx + (a_2x + b_2x + c_2)\,dy = 0$$

The idea is to transform this eqn to a form we can solve: homogeneous, exact, etc

We notice that:

(i) When $b_2 = 0$, the eqn is linear.

(ii) When $b_1 = b_2$, the eqn is exact.

(iii) When $a_1 b_2 = a_2 b_1$, the eqn is of the form $y' = f(ax + by)$.

Writing the eqn as

$$y' = -\frac{a_1x + b_1y + c_1}{a_2x + b_2y + c_2},$$ we

see that when $c_1 = c_2 = 0$, the eqn is homogeneous.

In fact, in general, we can transform this eqn into the homogeneous case, as long as $a_1 b_2 \neq a_2 b_1$.

To do this, notice that replacing x, y by $x + c, y + c'$ leave the LHS unchanged, and produce extra constants on both the nominator & denominator of RHS.

Inspired by this, we let

$x = u + h$, $y = v + k$, when v is the new unknown, u is the new variable and h, k are constants.

The eqn becomes:

$$\frac{dv}{du} = \frac{a_1 u + b_1 v + [a_1 h + b_1 k + c_1]}{a_2 u + b_2 v + [a_2 h + b_2 k + c_2]}$$

Now, we see that as long as we can find h, k satisfying

$$a_1 h + b_1 k + c_1 = 0$$
$$a_2 h + b_2 k + c_2 = 0$$

then our eqn is transformed into the homogeneous case. As $a_1 b_2 \neq a_2 b_1$, such h, k exist and are unique.

e.g.

(i) $(-3x + y - 1)\,dx + (x + y + 3)\,dy = 0$

Soln:

Here $a_1 = -3$, $b_1 = 1$, $a_2 = 1$, $b_2 = 1$

$a_1 b_2 = (-3)(1) \neq a_2 b_1 = (1)(1) = 1$.

∴ We need to solve:

$$-3h + k - 1 = 0$$
$$h + k + 3 = 0$$

to get $(h, k) = (-1, -2)$

$\therefore \quad x = u-1, \quad y = v-2 \quad$ with $\quad \cdots (5)$

$$\frac{dv}{du} = \frac{-3u+v}{u+v} = \frac{-3+\frac{v}{u}}{1+\frac{v}{u}} \quad \text{(homogeneous)}$$

Now let $v = uw \Rightarrow v' = w + uw'$

then eqn (5) becomes!

$$w + uw' = \frac{-3+w}{1+w}$$

$$\Rightarrow uw' = \frac{-3+w-w-w^2}{1+w} = \frac{-3-w^2}{1+w}$$

$$\vdots$$

$$(3x+y+5)(-x+y+1) = c.$$

(ii) Solve $(2x-y)\,dx + (4x+y-3)\,dy = 0$

Soln: $y' = -\frac{2x-y}{4x+y-3}$

Let $x = u+h$ and $y = v+k$

Thus we need to find $h, k \Rightarrow$

$$2h-k=0,$$
$$4h+k-3=0$$

$\therefore (h,k) = (\frac{1}{2}, 1)$

Next we solve: $\frac{dv}{du} = -\frac{2u-v}{4u+v}$

Now let $v = zu \Rightarrow v' = z + uz'$

$$\Rightarrow \quad z + uz' = -\frac{2-z}{4+z}$$

$$\Rightarrow \quad -\frac{4+z}{z^2+3z+2}\,dz = \frac{du}{u}$$

and $$-\frac{4+z}{z^2+3z+2} = \frac{-3}{z+1} + \frac{2}{z+2}$$

$$\therefore \quad -3\ln|z+1| + 2\ln|z+2| = \ln|u| + c$$

$$\Rightarrow \quad (z+2)^2 = e^c (z+1)^3 u$$

$$\Rightarrow \quad (y+2x-2)^2 = c_1 (2y+2x-3)^3$$

Note the above formula includes the case when $z = -2$. The case $z = -1$ corresponds to $v = -u$ or $y = \frac{3}{2} - x$

$\therefore$ The solns are:
$y = \frac{3}{2} - x$, $(y+2x-2)^2 = c_1(2y+2x-3)^3$.

(d) Exact eqns:

Defn:

The differential form $M(x,y)\,dx + N(x,y)\,dy$ is said to be exact in a rectangle R if there is a fn $F(x,y)$ $\ni$

$$\frac{\partial F(x,y)}{\partial x} = M(x,y) \quad \text{and} \quad \frac{\partial F(x,y)}{\partial y} = N(x,y)$$

$\forall\, x \in R$.

That is, the total differential of $F(x,y)$ satisfies

$$dF(x,y) = M(x,y)\,dx + N(x,y)\,dy.$$

If $M(x,y)\,dx + N(x,y)\,dy$ is exact differential form, then the eqn $M(x,y)\,dx + N\,dy = 0$ is called an exact eqn.

Test for exactness:

Theorem]:

Suppose the first partial derivatives of $M(x,y)$ & $N(x,y)$ are continuous in a Rectangle R. Then

$M\,dx + N\,dy = 0$ is an exact eqn in R iff the compatibility condition

$$\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$ holds for all $(x,y) \in R$.

e.g. $(2xy - \sec^2 x)\,dx + (x^2 + 2y)\,dy = 0$ ..... (5)

soln: $M(x,y) = 2xy - \sec^2 x$ and $N(x,y) = x^2 + 2y$

$$\frac{\partial M}{\partial y} = 2x = \frac{\partial N}{\partial x}$$

∴ eqn (5) is exact.

This implies $\exists$ a fn $F(x,y) \Rightarrow$

$$\frac{\partial F}{\partial x} = M \Rightarrow F = \int M\,dx + g(y)$$

$$\Rightarrow F = \int (2xy - \sec^2 x)\,dx + g(y)$$

$$= x^2 y - \tan(x) + g(y)$$

Also, $\frac{\partial F}{\partial y} = N$

$$\Rightarrow x^2 + g'(y) = x^2 + 2y$$

$$\Rightarrow g'(y) = 2y \Rightarrow g(y) = y^2 + c_1$$

$\therefore F(x,y) = x^2 y - \tan x + y^2 + c_1$ and the soln is

$$x^2 y - \tan x + y^2 = C.$$

(e) Linear first order eqns:

A differential eqn of the first order, which is linear is of the form:

$a(x)y' + b(x)y = f(x)$, which can be written in form

$$y' + P(x)y = Q(x) \quad \cdots - (6)$$

with $P(x) = \frac{b(x)}{a(x)}$ and $Q(x) = \frac{f(x)}{a(x)}$, $a(x) \neq 0$

One method of solving eqn (6) is by multiplying by $\mu(x)$ (an integrating factor).

This gives us

$$\mu y' + \mu P y = \mu Q$$

and then impose the upon μ the condition that

$$\mu y' + \mu P y = \frac{d}{dx}(\mu y)$$

$$\Rightarrow \frac{d\mu}{dx} = \mu P \Rightarrow \frac{d\mu}{\mu} = P(x)\,dx$$

$$\Rightarrow \ln|\mu| = \int P(x)\,dx + C$$

$$\Rightarrow \mu = \pm C\, e^{\int P\,dx}$$

Since we don't require the most general form of μ, we take $\pm C = 1$.

$$\Rightarrow \mu = e^{\int P\,dx}.$$

$\mu(x)$ is called an integrating factor

e.g.:

i) Find the general soln of

$$\frac{1}{x}\frac{dy}{dx} - \frac{2y}{x^2} = x\cos(x), \quad x > 0$$

soln:

$$\frac{1}{x}\frac{dy}{dx} - \frac{2y}{x^2} = x\cos(x)$$

$$\Rightarrow y' - \frac{2}{x}y = x^2\cos(x); \quad \cdots (7)$$

Here $P(x) = -\frac{2}{x}$, $Q(x) = x^2\cos(x)$

thus, $\mu(x) = e^{\int -\frac{2}{x}dx} = e^{-2\ln|x|} = x^{-2}$

Multiplying (7) by $\mu(x)$ we get

$$x^{-2}y' - 2x^{-3}y = \cos(x)$$

$$\Rightarrow \quad \frac{d}{dx}(x^{-2}y) = \cos(x)$$

$$\Rightarrow \quad x^{-2}y = \sin(x) + C$$

(iii) solve $y' + 3xy = 5^{-x}$

$$\mu(x) = e^{\int 3x\,dx} = e^{\frac{3}{2}x^2}$$

$$\therefore \quad e^{\frac{3}{2}x^2}y' + 3xe^{\frac{3}{2}x^2}y = 5^{-x}e^{\frac{3}{2}x^2}$$

$$\Rightarrow \quad \left(e^{\frac{3}{2}x^2}y\right)' = 5^{-x}e^{\frac{3}{2}x^2}$$

$$\Rightarrow e^{\frac{3}{2}x^2}y = \frac{5^{-x}}{3}e^{\frac{3}{2}x^2} + C$$

$$\Rightarrow y = \frac{5^{-x}}{3} + Ce^{-\frac{3}{2}x^2}.$$

(f) Bernoulli eqn:

A first order differential eqn that can be written in the form

$$y' + P(x)y = Q(x)y^n$$

is called the Bernoulli eqn. If $n=0$ it becomes linear first order. If $n=1$ it becomes separable eqn.

Solving the Bernoulli eqn with $n \neq 0, 1$:

Soln: $y' + P(x)y = Q(x)y^n$ ..... (8)

$\Rightarrow y^{-n}y' + Py^{-n+1} = Q(x)$

Let $z = y^{-n+1} \Rightarrow z' = (1-n)y^{-n}y'$

$\therefore$ eqn (8) becomes

$$\frac{z'}{1-n} + Pz = Q(x)$$

$$\Rightarrow z' + P(1-n)z = Q(x)$$

which is a linear eqn in z.

exercise: (Quiz)

Riccati eqn:

An eqn of the form

$$y' = P(x)y^2 + Q(x)y + R(x) \quad \text{---} (9)$$

is called a generalized Riccati eqn.

(a) If one soln - say, $u(x)$ of (9) is known, show that the substitution $y = u + \frac{1}{v}$ reduces eqn (9) to a linear eqn in v.

(b) Given that $u(x) = x$ is a solution to $y' = x^3(y-x)^2 + \frac{y}{x}$, use result of part

is to find all other solns to this eqn.

If Reducible to exact eqns:

If the eqn $M\,dx + N\,dy = 0$ is neither exact nor linear; it is reducible to exact if we can find a fn ρ $\ni$ $\rho M\,dx + \rho N\,dy = 0$ is exact, i.e. we want $\rho \ni$

$$\frac{\partial}{\partial y}(\rho M) = \frac{\partial}{\partial x}(\rho N)$$

$$\Rightarrow \quad \rho \frac{\partial M}{\partial y} + M \frac{\partial \rho}{\partial y} = \rho \frac{\partial N}{\partial x} + N \frac{\partial \rho}{\partial x}$$

$$\Rightarrow \quad \rho \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = N \frac{\partial \rho}{\partial x} - M \frac{\partial \rho}{\partial y}.$$

Suppose that ρ is a fn of x only, then

$$\rho \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) = N \frac{\partial \rho}{\partial x}$$

$$\Rightarrow \quad \frac{1}{N} \left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x} \right) dx = \frac{d\rho}{\rho} = f(x)$$

$$\Rightarrow \quad \rho = e^{\int f(x)\,dx}$$

Similarly, if ρ is a fn of y only, then

$$\rho = e^{-\int g(y)\,dy}$$

So we say

(i) if $\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = f(x)$, then

$$\rho = e^{\int f(x)\,dx}$$ is an integrating factor

(ii) if $\frac{1}{M}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = g(y)$, then

$$\rho = e^{-\int g(y)\,dy}$$ is an integrating factor.

e.g.

(i) Solve $(3x^2+y)\,dx + (x^2y - x)\,dy = 0$ ... (10)

$$\frac{\partial M}{\partial y} = 1 \neq \frac{\partial N}{\partial x} = 2xy - 1$$

Thus, the eqn is not exact.

But $\frac{1}{N}\left(\frac{\partial M}{\partial y} - \frac{\partial N}{\partial x}\right) = \frac{1}{x^2y - x}\left(1 - (2xy - 1)\right)$

$$= \frac{2 - 2xy}{x(xy-1)} = \frac{-2(xy-1)}{x(xy-1)}$$

$$= \frac{-2}{x} = f(x)$$

$\therefore \exists\ \rho \Rightarrow \rho = e^{\int -\frac{2}{x}\,dx} = x^{-2}$

Multiplying (10) by ρ we get:

$$(3 + x^{-2}y)\,dx + (y - x^{-1})\,dy = 0 \quad \ldots (11)$$

$$\frac{\partial}{\partial y}(3 + x^{-2}y) = x^{-2} = \frac{\partial}{\partial x}(y - x^{-1}) \Rightarrow \text{eqn is exact.}$$

Part 2:

2.1

2nd - order differential eqn is an eqn of the form $F(x, y, y', y'') = 0$

e.g.

i. $y'' = x^3$

ii. $xy'' - \cos(y) = e^{3x}$

iii. $y'' - 4xy' + y = 2$

The linear 2nd - order differential eqn has the form:

$$R(x)y'' + P(x)y' + Q(x)y = S(x)$$

Theorem 2.1

Let p, q and f be continuous on an open interval I. Let x_0 be in I and let A and B be any real #s. Then the initial value problem (2.1)

$$y'' + p(x)y' + q(x)y = f(x);$$

$y(x_0) = A$ and $y'(x_0) = B$ has a unique soln defined $\forall\, x \in I$.

2.2 The Homogeneous eqn:

If $f(x) = 0$ then eqn (2.1) is called the homogeneous eqn, i.e.

$$y'' + p(x)y' + q(x)y = 0 \quad \cdots \quad (2.2)$$

Theorem 2.2: Superposition principle or Linearity principle:

For a homogeneous linear ODE, any linear combination of two solns on an open interval, I, is again a soln. In particular, for such an eqn, sums and constant multiple of solns are solns.

Proof:

Let y_1 and y_2 be solns of eqn (2.2) on I

Then by substituting $y = c_1y_1 + c_2y_2$, we get:

$$y'' + py' + qy = (c_1y_1 + c_2y_2)'' + p(c_1y_1 + c_2y_2)' + q(c_1y_1 + c_2y_2)$$

$$= c_1y_1'' + c_1py_1' + c_1qy_1 + c_2y_2'' + pc_2y_2' + qc_2y_2$$

$$= c_1(y_1'' + py_1' + qy_1) + c_2(y_2'' + py_2' + qy_2)$$

$$= c_1(0) + c_2(0) = 0.$$

$\therefore y = c_1y_1 + c_2y_2$ is a soln if y_1 and y_2 are solns.

Defn: Linear dependence, Independence

Two fns f and g are linearly dependent on an open interval I if, for some constants c, either $f(x) = cg(x)$ $\forall x \in I$, or $g(x) = cf(x)$ $\forall x \in I$. If f & g are not linearly dependent on I, then they are said to be linearly independent on the interval.

There is a simple test to tell whether two solns of eqn (2.1) are linearly independent on an interval.

Define the Wronskian of solns y_1 and y_2 to be

$$W(x) = y_1(x)y_2'(x) - y_1'(x)y_2(x)$$

$$= \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix}$$

Defn:

Let y_1 and y_2 be solns of $y'' + py' + qy = 0$ on I.

1. y_1 and y_2 form a fundamental set of

of solns on I if y_1 and y_2 are linearly independent on I.

2. When y_1 and y_2 form a fundamental set of solns, we call $c_1y_1 + c_2y_2$, with c_1 and c_2 arbitrary constants, the general soln of the differential eqn on I.

2.3 The nonhomogeneous eqn:

$$y'' + p(x)y' + q(x)y = f(x) \quad \cdots (2.3)$$

Theorem 2.3:

Let y_1 and y_2 be a fundamental set of solns of $y'' + p(x)y' + q(x)y = 0$ on an open interval I.

Let y_p be any soln of eqn (2.3) on I.

Then, for any soln ψ of eqn (2.3), there exist numbers c_1 and c_2 $\ni$

$$\psi = c_1y_1 + c_2y_2 + y_p$$

This leads to call $c_1y_1 + c_2y_2 + y_p$ the general soln of eqn (2.3) and suggests that:

To solve $y'' + py' + qy = f(x)$;

1. find the general soln $c_1 y_1 + c_2 y_2$ of the homogeneous case.
2. find any soln y_p of $y'' + py' + qy = f(x)$, and
3. write the general soln $c_1 y_1 + c_2 y_2 + y_p$.

2.4. Solving 2nd-order ODEs (linear)

2.4.1 Reduction of order:

Given $y'' + py' + qy = 0$ . . . . . (2.4.1)

Suppose we know a soln $y_1 \neq 0$. We will look for a 2nd soln of the form $y_2(x) = u(x) y_1(x) \Rightarrow y_2' = u'y_1 + uy_1'$

and $y_2'' = u''y_1 + 2u'y_1' + uy_1''$

In order for y_2 to be a soln we need

$$u''y_1 + 2u'y_1' + uy_1'' + p(u'y_1 + uy_1') + q(uy_1) = 0$$

$$\Rightarrow u''y_1 + u'[2y' + py_1] + u[y_1'' + py_1' + qy_1] = 0$$

($y_1'' + py_1' + qy_1 \to 0$)

$$\Rightarrow u''y_1 + u'[2y' + py_1] = 0$$

On any interval in which $y_1 \neq 0$, we have

$$u'' + \frac{2y_1' + py_1}{y_1} u' = 0$$

Let $g(x) = \frac{2y_1' + py_1}{y_1} \Rightarrow u'' + g(x)u' = 0$

Let $v = u' \Rightarrow v' = u''$

$\therefore \quad v' + g(x)v = 0$

This is a 1st-order differential eqn for v, with the general soln

$$v(x) = c e^{-\int g(x)\,dx}$$

Since we need only one 2nd soln y_2, we choose $c = 1 \Rightarrow v(x) = e^{-\int g(x)\,dx}$

Finally, since $v = u'$

$$u(x) = \int \left(e^{-\int g(x)\,dx} \right) dx$$

Once $u(x)$ is obtained, then $y_2 = u y_1$.

e.g.

1. Suppose we are given $y_1 = e^{-2x}$ is one soln of $y'' + 4y' + 4y = 0$. To find the 2nd soln, let $y_2 = u(x) e^{-2x}$

$\Rightarrow y_2' = u' e^{-2x} - 2e^{-2x} u$ and $y_2'' = u'' e^{-2x} + 4e^{-2x} u - 4u' e^{-2x}$.

Subst. tute these into the differential

to get:

$$u''e^{-2x} + 4e^{-2x}u - 4u'e^{-2x} + 4(u'e^{-2x} - 2e^{-2x}u) + 4ue^{-2x} = 0$$

$$\Rightarrow \quad u''e^{-2x} = 0 \quad \text{or} \quad u'' = 0$$

$$\Rightarrow u(x) = cx + d$$

Since we only need one 2nd soln y_2, we choose $c = 1$ and $d = 0$.

$$\Rightarrow y_2(x) = xe^{-2x}.$$

The general soln is

$$y(x) = c_1 e^{-2x} + c_2 x e^{-2x}$$

2) Suppose we want the general soln of $y'' - \left(\frac{3}{x}\right)y' + \left(\frac{4}{x^2}\right)y = 0$ for $x > 0$, and somehow we find one soln $y_1 = x^2$.

Let $y_2 = x^2 u \Rightarrow y_2' = 2xu + x^2u'$ and

$$y_2'' = 2u + 4xu' + x^2u''$$

Substitute into the differential eqn to get

$$2u + 4xu' + x^2u'' - \frac{3}{x}(2xu + x^2u') + \frac{4}{x^2}(x^2u) = 0$$

$$\Rightarrow \quad x^2u'' + xu' = 0$$

$$\Rightarrow \quad xu'' + u' = 0 \quad (x > 0)$$

with $v = u' \Rightarrow xv' + v = (xv)' = 0 \Rightarrow xv = c$

We choose $c = 1$. Then

$$v = u' = \frac{1}{x}$$

$$\Rightarrow u = \ln(x) + d = \ln(x) \quad (d = 0)$$

$$\therefore y_2 = x^2 \ln(x)$$

and $$y = c_1 x^2 + c_2 x^2 \ln(x)$$

2.4.2 The constant coefficient homogeneous linear eqn:

The linear homogeneous eqn

$$y'' + Ay' + By = 0 \quad \cdots \quad (2.4.2)$$

in which A and B are #s, occurs frequently in important applications.

There is a standard approach to solving this eqn.

Let $y(x) = e^{\lambda x} \Rightarrow \lambda^2 e^{\lambda x} + A\lambda e^{\lambda x} + Be^{\lambda x} = 0$

$$\Rightarrow \lambda^2 + A\lambda + B = 0$$

This is called the characteristic eqn.

Its roots are

$$\lambda = \frac{-A \pm \sqrt{A^2 - 4B}}{2},$$

leading to three cases:

i) $A^2 - 4B > 0$

In this case the characteristic eqn has two real, distinct roots,

$$a = -\frac{A + \sqrt{A^2-4B}}{2} \quad \text{and} \quad b = -\frac{A - \sqrt{A^2-4B}}{2}$$

yielding solns $y_1(x) = e^{ax}$ and $y_2(x) = e^{bx}$.

$$\therefore y = c_1 e^{ax} + c_2 e^{bx}$$

e.g. The characteristic eqn of

$$y'' - y' - 6y = 0 \text{ is } \lambda^2 - \lambda - 6 = 0$$

$$\Rightarrow \lambda_1 = -2 \text{ and } \lambda_2 = 3$$

$$\Rightarrow y = c_1 e^{-2x} + c_2 e^{3x}$$

ii) $A^2 - 4B = 0$

Now the characteristic eqn has the repeated root $\lambda = -\frac{A}{2}$, so $y_1(x) = e^{-\frac{A}{2}x}$ is one soln.

This method doesn't provide the second soln. But using the reduction of order gives us $y_2(x) = xe^{-\frac{A}{2}x}$

$$\therefore y(x) = c_1 e^{-\frac{Ax}{2}} + c_2 e^{-Ax/2} = e^{-Ax/2}(c_1 + c_2 x)$$

e.g. The characteristic eqn of

$y'' - 6y' + 9y = 0$ is $\lambda^2 - 6\lambda + 9 = 0$, with $\lambda = 3$ (repeated)

$$\therefore \quad y(x) = e^{3x}(c_1 + c_2 x).$$

(iii) $A^2 - 4B < 0$

Now the characteristic has complex roots:

$$\frac{-A \pm \sqrt{4B - A^2}\, i}{2}$$

Let $p = -\frac{A}{2}$ and $q = \frac{1}{2}\sqrt{4B - A^2}$

so the roots are $p \pm iq$. This yields two solns

$$y_1(x) = e^{(p+iq)x} \text{ and } y_2(x) = e^{(p-iq)x}$$

$$\Rightarrow y(x) = c_1 e^{(p+iq)x} + c_2 e^{(p-iq)x}$$

Remember the Euler formula:

$$e^{i\theta} = \cos\theta + i\sin\theta$$

$$\Rightarrow e^{iqx} = \cos(qx) + i\sin(qx)$$

$$\& \; e^{-iqx} = \cos(qx) - i\sin(qx)$$

$$\Rightarrow y = c_1 e^{(p+iq)x} + c_2 e^{(p-iq)x}$$

$$= c_1 e^{px} \cdot e^{iqx} + c_2 e^{px} \cdot e^{-iqx}$$

$$= c_1 e^{px}(\cos(qx) + i\sin(qx)) + c_2 e^{px}(\cos(qx) - i\sin(qx))$$

$$= e^{px}[(c_1 + c_2)\cos(qx) + (ic_1 - ic_2)\sin(qx)]$$

$\Rightarrow y = e^{px}(c_3\cos(qx) + c_4\sin(qx))$

e.g.

1) Solve $y'' + 4y' + 7y = 0$

Soln:

$\lambda^2 + 4\lambda + 7 = 0 \Rightarrow \lambda_1 = -2 + \sqrt{3}i$

and $\lambda_2 = -2 - \sqrt{3}i$

$\therefore y = e^{-2x}(c_1\cos(\sqrt{3}x) + c_2\sin(\sqrt{3}x))$

2.4.3 Method of undetermined coeff:

The underlying idea behind this method is a conjecture about y_p, an educated guess.

The method is limited to the f.f. types of y_p and their linear combinations:

$P(x) = a_n x^n + a_{n-1}x^{n-1} + \cdots + a_1 x + a_0$, $P(x)e^{ax}$, $P(x)e^{ax}\sin\beta x$ and $P(x)e^{ax}\cos\beta x$, where n is nonnegative integer and $\alpha, \beta \in \mathbb{R}$.

This method is not applicable to $g(x) = \ln(x)$, $g(x) = \frac{1}{x}$, $g(x) = \tan x$, $g(x) = \sin^{-1}x$ and so on, which will be

handled by the method in next section.

Table 2.4.3: Method of Undetermined coefficients:

Term in $g(x)$	Choice of $y_p(x)$
$ke^{\gamma x}$	$Ce^{\gamma x}$
kx^n $(n=0,1,\dots)$	$K_n x^n + K_{n-1}x^{n-1} + \dots + K_1 x + K_0$
$k\cos(\omega x)$ $k\sin(\omega x)$	$k\cos(\omega x) + M\sin(\omega x)$
$ke^{\alpha x}\cos(\omega x)$ $ke^{\alpha x}\sin(\omega x)$	$e^{ax}(k\cos(\omega x) + M\sin(\omega x))$

Choice Rules for the method of undetermined coefficients:

(a) If $g(x)$ is one of the fns in 1st column in table 2.4.3, choose y_p in the same line.

(b) If a term in your choice of y_p appears in y_c, multiply your choice of of y_p by x (or by x^2 if this soln is a double root)

e.g.

Solve $y'' + 4y' - 2y = 2x^2 - 3x + 6$

soln: $y_c = C_1 e^{-(2+\sqrt{6})x} + C_2 e^{(-2+\sqrt{6})x}$

$\therefore y_p = Ax^2 + Bx + C$

$\Rightarrow y_p' = 2Ax + B \Rightarrow y_p'' = 2A$

$\therefore y_p'' + 4y_p' - 2y_p = 2A + 8Ax + 4B - 2Ax^2 - 2Bx - 2C$

$= 2x^2 - 3x + 6$

Comparing coefficients:

$-2A = 2, \Rightarrow A = -1$

$8A - 2B = -3 \Rightarrow -2B = -3 - 8(-1) = 5 \Rightarrow B = -\frac{5}{2}$

$2A + 4B - 2C = 6 \Rightarrow C = -9$

$\therefore y_p = -x^2 - \frac{5}{2}x - 9$

$\Rightarrow y = y_c + y_p = c_1 e^{-(2+\sqrt{6})x} + c_2 e^{(-2+\sqrt{6})x} - x^2 - \frac{5}{2}x - 9$

b) $y'' + 3y' + 2y = 30e^{2x}$

Soln: For y_c:

$\lambda^2 + 3\lambda + 2 = 0$

$\Rightarrow (\lambda + 1)(\lambda + 2) = 0 \Rightarrow \lambda = -1, -2$

$\therefore y_c = c_1 e^{-x} + c_2 e^{-2x}$

$y_p = Ae^{2x} \Rightarrow y_p' = 2Ae^{2x}$ and $y_p'' = 4Ae^{2x}$

$\therefore y_p'' + 3y_p' + 2y_p = 4Ae^{2x} + 3(2Ae^{2x}) + 2Ae^{2x} = 30e^{2x}$

$\Rightarrow 12Ae^{2x} = 30e^{2x}$

$\Rightarrow 12A = 30 \Rightarrow A = \frac{30}{12} = \frac{5}{2}$

$\therefore y_p = \frac{5}{2}e^{2x} \Rightarrow y = c_1 e^{-x} + c_2 e^{-2x} + \frac{5}{2}e^{2x}$

(c) solve $y'' - 2y' = 4$

Soln: In this case, $y_c = A + Be^{2x}$

If we try $y_p = C \Rightarrow y_p'' - 2y_p = 0 \stackrel{?}{=} 4$ (contradiction)

To deal with this problem, we multiply our "guess" by an extra x to avoid duplication of solns:

$\therefore y_p = Cx \Rightarrow -2C = 4 \Rightarrow C = -2$

$\therefore y_p = -2x$ and $y = A + Be^{2x} - 2x$

(d) Solve $y'' = 4$

Soln: $y_c = Ax + B$, (repeated root, $\lambda = 0$)

$\therefore y_p = Cx^2 \Rightarrow y_p' = 2Cx$ and $y_p'' = 2C = 4$

$\Rightarrow C = 2$

$\therefore y_p = 2x^2$ and $y = y_c + y_p$

$= Ax + B + 2x^2$

(e) Solve $y'' + y - 6y = 5e^{-3x}$

Soln: $y_c = Ae^{2x} + Be^{-3x}$

Because e^{-3x} is a soln of y_c, we try $y_p(x) = Cxe^{-3x} \Rightarrow y_p' = Ce^{-3x} - 3Cxe^{-3x}$

and $y_p'' = -3Ce^{-3x} - 3Ce^{-3x} + 9Cxe^{-3x} = -6Ce^{-3x} + 9Cxe^{-3x}$

So $y_p'' + y' - 6y = -6Ce^{-3x} + 9Cxe^{-3x} + Ce^{-3x} - 3Cxe^{-3x}$

$-6Cxe^{-3x} = 5e^{-3x}$

$\Rightarrow C = -1$ and $y_p = -xe^{-3x}$

$\therefore y = y_c + y_p = Ae^{2x} + Be^{-3x} - xe^{-3x}$.

(f) Solve $y'' + 2y' + y = 6e^{-x}$

Soln: $y_c = Ae^{-x} + Bxe^{-x} \Rightarrow y_p = Cx^2e^{-x}$

Substituting y_p in the above eqn, we get $C = 3 \Rightarrow y_p = 3x^2e^{-x}$

$\therefore y = y_c + y_p = Ae^{-x} + Bxe^{-x} + 3x^2e^{-x}$
$= (A + Bx + 3x^2)e^{-x}$

(g) Solve: $y'' + y = 8\cos(x)$

Soln: $y_c = A\sin(x) + B\cos(x) \Rightarrow y_p = Cx\sin(x) + Dx\cos(x)$

Substituting y_p, y_p', y_p'' in the above eqn, we get:

$C = 4$ and $D = 0 \Rightarrow y_p = 4x\sin(x)$

$\therefore y = y_c + y_p = (A + 4x)\sin(x) + B\cos(x)$.

2.4.4 Variation of parameters:

Now, let's consider the 2nd-order ODE of the form:

$$y'' + P(x)y' + Q(x)y = f(x) \quad \cdots (2.4.4)$$

We have already shown that the

complementary soln: y_c will take the form $y_c = c_1 y_1 + c_2 y_2$. Analogous to this, we now ask: Can the parameters c_1 and c_2 in y_c be replaced with fns $u_1(x)$ and $u_2(x)$ so that

$$y_p(x) = u_1 y_1 + u_2 y_2 \quad \text{is a particular soln?}$$

To answer this question, Let's find y_p, y_p' and y_p'' and substitute in eqn (2.4.4) above:

$$y_p' = u_1 y_1' + y_1 u_1' + u_2 y_2' + y_2 u_2' \quad \text{and}$$

$$y_p'' = u_1 y_1'' + 2y_1' u_1' + y_1 u_1'' + u_2 y_2'' + 2u_2' y_2' + y_2 u_2''$$

$$\therefore y_p'' + P y_p' + Q y_p = u_1 [y_1'' + P y_1' + Q y_1] + u_2 [y_2'' + P y_2' + Q y_2]$$

(y_1 soln: $= 0$; y_2 soln: $= 0$)

$$+ y_1 u_1'' + u_1' y_1' + y_2 u_2'' + u_2' y_2' + P[y_1 u_1' + y_2 u_2'] + y_1' u_1' + y_2' u_2'$$

$$= \frac{d}{dx}[y_1 u_1' + y_2 u_2'] + P[y_1 u_1' + y_2 u_2'] + y_1' u_1' + y_2' u_2' = f(x) \quad (2.4.4b)$$

We need two eqns to solve for u_1 and u_2. We need to assume that

$$y_1 u_1' + y_2 u_2' = 0 \quad \text{so that} \quad y_1' u_1' + y_2' u_2' = f(x)$$

in (2.4.4b) ... we need to solve:

$$y_1 u_1' + y_2 u_2' = 0$$
$$y_1' u_1' + y_2' u_2' = f(x) \quad \text{for } u_1 \text{ & } u_2.$$

Using Cramer's rule, the soln of the system is

$$u_1' = \frac{W_1}{W} = \frac{-y_2 f(x)}{W} \quad \text{and} \quad \Rightarrow u_1 = -\int \frac{y_2 f(x)}{W} dx + c$$

$$u_2' = \frac{W_2}{W} = \frac{y_1 f(x)}{W}, \quad \text{where} \quad \Rightarrow u_2 = \int \frac{y_1 f(x)}{W} dx$$

$$W = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix}, \quad W_1 = \begin{vmatrix} 0 & y_2 \\ f(x) & y_2' \end{vmatrix}, \quad W_2 = \begin{vmatrix} y_1 & 0 \\ y_1' & f(x) \end{vmatrix}$$

Examples:

(a) Let's revisit example (e) in the previous section: $y'' + y - 6y = 5e^{-3x}$

Soln: $y_c = c_1 e^{2x} + c_2 e^{-3x} \Rightarrow$

$$y_p = u_1 e^{2x} + u_2 e^{-3x}$$

$$W(e^{2x}, e^{-3x}) = \begin{vmatrix} e^{2x} & e^{-3x} \\ 2e^{2x} & -3e^{-3x} \end{vmatrix} = -3e^{-x} - 2e^{-x} = -5e^{-x}$$

$$W_1 = \begin{vmatrix} 0 & e^{-3x} \\ 5e^{-3x} & -3e^{-3x} \end{vmatrix} = -5e^{-6x}$$

$$W_2 = \begin{vmatrix} e^{2x} & 0 \\ 2e^{2x} & 5e^{-3x} \end{vmatrix} = 5e^{-x}$$

$$\therefore u_1 = -\int \frac{y_2 f(x)}{W} dx = +\int \frac{e^{-3x} \cdot 5e^{-3x}}{+5e^{-x}} dx = \int \frac{e^{-6x}}{e^{-x}} dx$$

$$= \int e^{-5x} dx = -\frac{1}{5} e^{-5x} + c_1$$

$$u_2 = \int \frac{y_1 f(x)}{W} dx = \int \frac{e^{2x} \cdot 5e^{-3x}}{-5e^{-x}} dx = -\int dx = -x + c_2$$

$$\therefore\ y_p = -\frac{1}{5}e^{-5x}\cdot e^{2x} + (-1)e^{-3x}$$

$$= -\frac{1}{5}e^{-3x} - xe^{-3x}$$

$$\therefore\ y = y_c + y_p = c_1e^{2x} + c_2e^{-3x} - \frac{1}{5}e^{-3x} - xe^{-3x}$$

$$= c_1e^{2x} + \left(c_2 - \frac{1}{5}\right)e^{-3x} - xe^{-3x}$$

$$= c_1e^{2x} + c_3e^{-3x} - xe^{-3x}$$

(equivalent soln found in the undermined coefficient case, please the soln to example (e, in the previous section.)

(b, Solve $y''' - 2y'' - y' + 2y = e^{4x}$

soln:

For y_c: $\lambda^3 - 2\lambda^2 - \lambda + 2 = 0 \Rightarrow (\lambda-2)(\lambda-1)(\lambda+1) = 0$

$\Rightarrow \lambda = -1, 1, 2$

$\therefore y_c = c_1e^{-x} + c_2e^{x} + c_3e^{2x} \Rightarrow$

$$y_p = u_1e^{-x} + u_2e^{x} + u_3e^{2x}$$

$$W = \begin{vmatrix} e^{-x} & e^{x} & e^{2x} \\ -e^{-x} & e^{x} & 2e^{2x} \\ e^{-x} & e^{x} & 4e^{2x} \end{vmatrix} = 6e^{2x},\quad W_1 = \begin{vmatrix} 0 & e^{x} & e^{2x} \\ 0 & e^{x} & 2e^{2x} \\ e^{4x} & e^{x} & 4e^{2x} \end{vmatrix} = e^{7x}$$

$$W_2 = \begin{vmatrix} e^{-x} & 0 & e^{2x} \\ -e^{-x} & 0 & 2e^{2x} \\ e^{-x} & e^{4x} & 4e^{2x} \end{vmatrix} = -3e^{5x}$$

$$W_3 = \begin{vmatrix} e^{-x} & e^{x} & 0 \\ -e^{-x} & e^{x} & 0 \\ e^{-x} & e^{x} & e^{4x} \end{vmatrix} = 2e^{4x}$$

$$\therefore\ u_1 = \int \frac{W_1}{W}\,dx = \int \frac{e^{7x}}{6e^{2x}}\,dx = \frac{1}{6}\int e^{5x}\,dx = \frac{1}{30}e^{5x}$$

$$u_2 = \int \frac{-3e^{5x}}{6e^{2x}}\,dx = -\frac{1}{2}\int e^{3x}\,dx = -\frac{1}{6}e^{3x}$$

$$u_3 = \int \frac{2e^{4x}}{6e^{2x}}\,dx = \frac{1}{3}\int e^{2x}\,dx = \frac{1}{6}e^{2x}$$

$$\therefore y_p = \frac{1}{30}e^{5x}\cdot e^{-x} + \left(-\frac{1}{6}\right)e^{3x}\cdot e^{x} + \frac{1}{6}e^{2x}\cdot e^{2x}$$

$$= \frac{1}{30}e^{4x} - \frac{1}{6}e^{4x} + \frac{1}{6}e^{4x}$$

$$= \frac{1}{30}e^{4x}$$

$$\therefore y = c_1 e^{-x} + c_2 e^{x} + c_3 e^{2x} + \frac{1}{30}e^{4x}$$

2.4.5 Cauchy-Euler eqn:

$$a_n x^n \frac{d^n y}{dx^n} + a_{n-1}x^{n-1}\frac{d^{n-1}y}{dx^{n-1}} + \cdots + a_1 x\frac{dy}{dx} + a_0 y = \underbrace{g(x)}$$

where $a_n, a_{n-1}, \ldots, a_0$ are constants.

For 2nd-order:

$$ax^2y'' + bxy' + cy = g(x)$$

Use the transformation: $y = x^m$, to get an auxiliary eqn and solve with cases as we did with homogeneous with constant coefficients.

Inverse operator method:

Given $f(D)y = R(x)$

General soln: $y = y_c + y_p$

$y_p = \frac{1}{f(D)} R(x)$, with $\frac{1}{f(D)}$ is the inverse operat

Case 1: $R(x) = e^{ax}$, $y_p = \frac{1}{f(D)} e^{ax} = \frac{1}{f(a)} e^{ax}$, $f(a) \neq 0$

If $f(a) = 0$, $y_p = \frac{1}{f(D)} e^{ax} = \frac{1}{(D-a)^n g(D)} e^{ax}$

$= \frac{x^n}{n!} \left(\frac{1}{g(a)} \cdot e^{ax} \right)$

where $(D-a)^n$ are factors that will make $f(D) = 0$ and $g(D)$ are other factors.

Case 2: $R(x)$ is either $\sin(ax)$ or $\cos(ax)$

A) $y_p = \frac{1}{f(D^2)} \sin(ax) = \frac{\sin(ax)}{f(-a^2)}$; $f(-a^2) \neq 0$

B) $y_p = \frac{1}{f(D^2)} \cos(ax) = \frac{\cos(ax)}{f(-a^2)}$; $f(-a^2) \neq 0$

C) $y_p = \frac{1}{D^2 + a^2} \sin(ax) = \frac{-x}{2a} \cos(ax)$; $f(-a^2) = 0$

D) $y_p = \frac{1}{D^2 + a^2} \cos(ax) = \frac{x}{2a} \sin(ax)$; $f(-a^2) = 0$

Case 3: $R(x)$ is a polynomial in x:

$y_p = \frac{1}{f(D)} x^n = \left(b_0 \pm b_1 D \pm b_2 D^2 \pm \cdots \pm b_n D^n \right) x^n$

e.g. $D(D+3)y = e^{3x} + 4x^2 + 3x$

Soln:

y_c: $\lambda(\lambda-3)=0 \Rightarrow \lambda=0,3$

$\Rightarrow y_c = c_1 + c_2 e^{3x}$

and $y_p = \frac{1}{D^2+3D}\{e^x + 4x^2+x\}$

$= \frac{1}{1^2+3(1)} e^x + \frac{1}{D^2+3D}(4x^2+x)$

$\frac{1}{D}\frac{1}{D+3} = \frac{1}{D}\cdot\frac{1}{3}\frac{1}{(1+\frac{D}{3})}$

$= \frac{1}{3D}\left(1+\frac{D}{3}\right)^{-1} = \frac{1}{3D}\left[1 - \frac{D}{3} + \frac{(-1)(-2)}{1\times 2}\left(\frac{D}{3}\right)^2 + \frac{(-1)(-2)(-3)}{1\times 2\times 3}\left(\frac{D}{3}\right)^3 + \right.$ n

$= \frac{1}{3D} - \frac{1}{9} + \frac{D}{27} - \frac{D^2}{81} + \cdots$

$\Rightarrow y_p = \frac{1}{4}e^x + \frac{1}{3}\left(\frac{4}{3}x^3+x^2\right) - \frac{1}{9}(4x^2+4) + \frac{1}{27}(8x+1)$

$- \frac{8}{81}$

3. Laplace transform:

Let f be a function defined for $t \geq 0$. Then the integral

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt$$

is said to be the Laplace transform of f, provided the integral converges.

Notation: $\mathcal{L}\{f(t)\} = F(s)$, $\mathcal{L}\{g(t)\} = G(s)$, $\mathcal{L}\{y(t)\} = Y(t)$.

Examples:

1. Evaluate $\mathcal{L}\{1\}$

Soln: $\mathcal{L}\{1\} = \int_0^\infty e^{-st}(1)\,dt = \lim_{b\to\infty} \int_0^b e^{-st}\,dt$

$$= \lim_{b\to\infty} \left[\frac{-e^{-st}}{s}\right]_0^b$$

$$= \lim_{b\to\infty} \frac{-e^{-st}+1}{s} = \frac{1}{s}$$

provided $s > 0$, since as $s \to \infty$, $e^{-st} \to 0$.

2. Find $\mathcal{L}\{t\}$:

Soln:

$$\mathcal{L}\{t\} = \int_0^\infty e^{-st} t\,dt = \frac{1}{s^2}$$

3 Evaluate (a) $\mathcal{L}\{e^{-t}\}$, (b) $\mathcal{L}\{e^{-2t}\}$, (c) $\mathcal{L}\{e^{st}\}$

Soln:

(a) $\mathcal{L}\{e^{-t}\} = \int_0^\infty e^{-t} e^{-st}\,dt = \int_0^\infty e^{-(s+1)t}\,dt$

$= \frac{1}{s+1}$, $s > -1$

(b) $\mathcal{L}\{e^{-2t}\} = \int_0^\infty e^{-2t} e^{-st}\,dt = \int_0^\infty e^{-(s+2)t}\,dt$

$= \frac{1}{s+2}$, $s > -2$

(c) $\mathcal{L}\{e^{st}\} = \int_0^\infty e^{st} e^{-st}\,dt$

$= \frac{1}{s-s}$

4. Find $\mathcal{L}\{\sin(3t)\}$

soln:

$\mathcal{L}\{\sin(3t)\} = \int_0^\infty \sin(3t)\cdot e^{-st}\,dt$

$= \frac{3}{s^2+9}$

Theorem 3.0a: Transforms of basic functions:

(a) $\mathcal{L}\{1\} = \frac{1}{s}$, (b) $\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}}$, $n = 1, 2, \ldots$

(c) $\mathcal{L}\{e^{at}\} = \frac{1}{s-a}$, (d) $\mathcal{L}\{\sin(kt)\} = \frac{k}{s^2+k^2}$

(e) $\mathcal{L}\{\cos kt\} = \frac{s}{s^2+k^2}$

(f) $\mathcal{L}\{\sinh kt\} = \frac{k}{s^2-k^2}$

(g) $\mathcal{L}\{\cosh kt\} = \frac{s}{s^2-k^2}$

(h) $\mathcal{L}\{e^{at}\sin(bt)\} = \frac{b}{(s-a)^2+b^2}$

(i) $\mathcal{L}\{e^{at}\cos(bt)\} = \frac{s-a}{(s-a)^2+b^2}$

(j) $\mathcal{L}\{t^n e^{at}\} = \frac{n!}{(s-a)^{n+1}}$

Piecewise continuous fn:

Evaluate $\mathcal{L}\{f(t)\}$ where $f(t) = \begin{cases} 0, & 0 \le t < 3 \\ 2, & t \ge 3 \end{cases}$

Soln:

$$\mathcal{L}\{f(t)\} = \int_0^\infty e^{-st} f(t)\,dt = \int_0^3 e^{-st}(0)\,dt + \int_3^\infty e^{-st}(2)\,dt$$

$$= 0 + \frac{2e^{-st}}{-s}\Big|_3^\infty$$

$$= \frac{2e^{-st}}{s}, \quad s > 0$$

Theorem 3.0b: s-shifting

If $f(t)$ has the transform $F(s)$ (where $s > k$ for some k), then $e^{at}f(t)$ has the transform $F(s-a)$ (where $s-a > k$). In

particular, $\mathcal{L}\{e^{at}f(t)\} = F(s-a)$

or if we take the inverse on both sides,

$$e^{at}f(t) = \mathcal{L}^{-1}\{F(s-a)\}$$

Proof:

$$F(s-a) = \int_0^\infty e^{-(s-a)t} f(t)\,dt$$

$$= \int_0^\infty e^{-st}\left[e^{at}f(t)\right]dt = \mathcal{L}\{e^{at}f(t)\}$$

Theorem 3.0c: Laplace transform of derivatives: $\mathcal{L}\{f'\} = \int_0^\infty e^{-st}f'(t)\,dt = e^{-st}f(t)\Big|_0^\infty + s\int_0^\infty e^{-st}f(t)\,dt$

1) $\mathcal{L}\{f'\} = s\mathcal{L}\{f\} - f(0) = sF - f(0)$

2) $\mathcal{L}\{f''\} = s^2\mathcal{L}\{f\} - sf(0) - f'(0)$

$$= s^2F - sf(0) - f'(0)$$

In general,

Let $f, f', \ldots, f^{(n-1)}$ be continuous for all $t \geq 0$. Furthermore, let $f^{(n)}$ be piecewise continuous on every finite interval on the semi-axis $t \geq 0$. Then the transform of $f^{(n)}$ satisfies

$$\mathcal{L}\{f^{(n)}\} = s^n\mathcal{L}\{f\} - s^{n-1}f(0) - s^{n-2}f'(0) - \cdots - f^{(n-1)}(0).$$

In transforms:

If $F(s)$ represents Laplace tranform of a fn $f(t)$, i.e., $\mathcal{L}\{f(t)\} = \mathcal{L}\{F(s)\}$ then inverse Laplace transform of $F(s)$ is given by:

$$f(t) = \mathcal{L}^{-1}\{F(s)\}$$

e.g.

$$\mathcal{L}\{1\} = \frac{1}{s} \Rightarrow 1 = \mathcal{L}^{-1}\left\{\frac{1}{s}\right\}$$

$$\mathcal{L}\{t\} = \frac{1}{s^2} \Rightarrow t = \mathcal{L}^{-1}\left\{\frac{1}{s^2}\right\}$$

$$\mathcal{L}\{e^{-3t}\} = \frac{1}{s+3} \Rightarrow e^{-3t} = \mathcal{L}^{-1}\left\{\frac{1}{s+3}\right\}$$

$$\mathcal{L}\{t^n\} = \frac{n!}{s^{n+1}} \Rightarrow t^n = \mathcal{L}^{-1}\left\{\frac{n!}{s^{n+1}}\right\}$$

Similarly,

$$\mathcal{L}^{-1}\left\{\frac{s}{s^2+k^2}\right\} = \cos kt, \quad \mathcal{L}^{-1}\left\{\frac{k}{s^2+k^2}\right\} = \sin kt$$

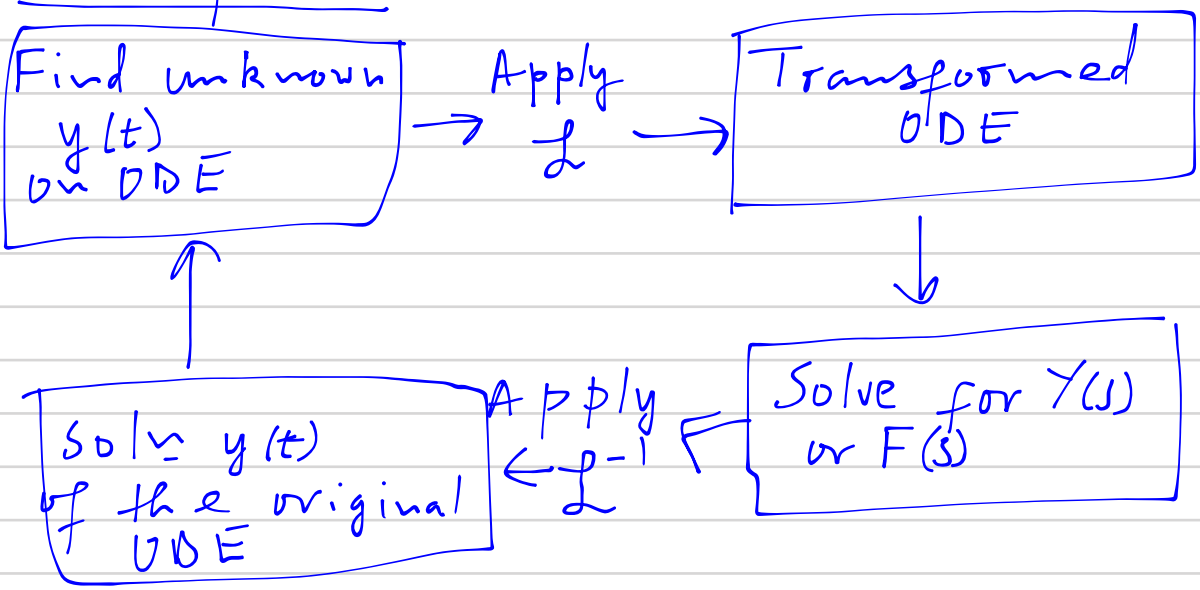
$$\mathcal{L}^{-1}\left\{\frac{1}{s^5}\right\} = \frac{1}{4!}\mathcal{L}^{-1}\left\{\frac{4!}{s^{4+1}}\right\} = \frac{1}{24}t^4$$

$$\mathcal{L}^{-1}\left\{\frac{1}{s^2+7}\right\} = \frac{1}{\sqrt{7}}\mathcal{L}^{-1}\left\{\frac{\sqrt{7}}{s^2+7}\right\} = \frac{1}{\sqrt{7}}\sin\sqrt{7}t$$

$$\mathcal{L}^{-1}\left\{\frac{s^2+6s+9}{(s-1)(s-2)(s+4)}\right\} = \mathcal{L}^{-1}\left\{\frac{A}{s-1} + \frac{B}{s-2} + \frac{C}{s+4}\right\}$$

← partial fractions

Solving ODEs with Laplace transforms:



Examples:

Part 4: Systems of differential eqns (linear)

A soln of a system of differential eqns is a set of sufficiently differentiable fns $x = \phi_1(t)$, $y = \phi_2(t)$, $z = \phi_3(t)$ and so on that satisfies each each eqn in the system of eqns on some interval I.

Methods of finding a soln to a system of ODEs:

4.1 Soln by elimination.

The method of elimination is used mostly in high school to solve a system of linear eqns. In the same manner, we will apply the elimination method to solve a system of ODEs.

e.g.

Consider $\dot{x} = \frac{dx}{dt} = -y + t$

$\dot{y} = x - t$

$$\Rightarrow \quad Dx = -y + t \quad \cdots\cdots (1)$$
$$Dy = x - t \quad \cdots\cdots (2)$$

$$\Rightarrow \quad Dx - y = t$$
$$Dy - x = -t$$

Applying D on the 1st eqn and keeping the 2nd as it is, we get

$$\left.\begin{aligned} D^2x - Dy &= 1 \\ Dy - x &= -t \end{aligned}\right\}$$ eliminating Dy from the 1st eqn

$$\Rightarrow \begin{cases} D^2x - (x - t) = 1 \end{cases}$$

$$\Rightarrow D^2x - x + t = 1$$
$$\Rightarrow D^2x - x = 1 - t,$$ which is a linear 2nd-order ODE, which can be solved using the methods from the previous section.

$$x_c(t) = c_1 e^{-t} + c_2 e^{t}$$

$$x_p(t) = A + Bt$$
$$x_p' = B \Rightarrow x_p'' = 0$$

$$\Rightarrow x_p'' - x_p = -(A + Bt) = 1 - t$$
$$\Rightarrow -A = 1 \Rightarrow A = -1$$
and $-B = -1 \Rightarrow B = 1$

$$\therefore x_p = t - 1$$

and $$x = x_c(t) + x_p(t) = c_1 e^{-t} + c_2 e^{t} + t - 1.$$

Now, make a back substitution to find $y(t)$.

2. Solve

$$\dot{x}(t) = 3x(t) - 4y(t) + 1,$$

$$\dot{y}(t) = 4x(t) - 7y(t) + 10t$$

Soln:

$$(D-3)x + 4y = 1$$

$$-4x + (D+7)y = 10t$$

We eliminate x by adding 4 times the 1st-eqn to $(D-3)$ applied to the 2nd eqn to get:

$$(16 + (D-3)(D+7))y = 4\cdot 1 + (D-3)(10t)$$
$$= 4 + 10 - 30t$$

$$\Rightarrow (D^2 + 4D - 5)y = 14 - 30t$$

This is a 2nd-order ODE which can be solved with methods learnt in the previous parts to get:

$$y(t) = c_1 e^{-5t} + c_2 e^{t} + 6t + 2$$

$x(t)$ can be found as follows:

Method 1: Do another elimination, but this time eliminate y, then solve the resulting ODE for $x = c_3 e^{-5t} + c_4 e^{t} + 8t + 5$

OR

Method 2: Substitute $y(t)$, $\dot{y}(t)$ into the

eqn which contains $x(t)$ and solve for $x(t)$.

3. Solve by elimination

(a) $\dot{x} + \dot{y} - x = s^-$

$\dot{x} + \dot{y} + y = 1$

(b) $\dot{x} = x - 4y$

$\dot{y} = x + y$

Homogenous linear systems:

The homogenous linear system of the 1st order ODE is given by

$$a_1\dot{\bar{X}} + b_1\bar{X} = 0, \quad \text{e.g.}$$

$\dot{\bar{X}} = \begin{pmatrix} 1 & 3 \\ 5 & 3 \end{pmatrix}\bar{X}$ has the generals soln of

$$\bar{X}_c = c_1\bar{X}_1 + c_2\bar{X}_2$$

The soln vectors $\bar{X}_1$ and $\bar{X}_2$ have the form

$$\bar{X}_i = \begin{pmatrix} k_1 \\ k_2 \end{pmatrix} e^{\lambda_i t}, \quad i = 1, 2,$$

where k_1, k_2, λ_1 and λ_2 are constants.

We now start to think that we will always find the soln of the form

$$\bar{X} = \begin{pmatrix} k_1 \\ k_2 \\ \vdots \\ k_n \end{pmatrix} e^{\lambda t} = \bar{k}e^{\lambda t}$$

for

$$\dot{\bar{X}} = A\bar{X}$$

where A is an $n \times n$ matrix of constants

Eigenvalues and Eigenvectors:

Given that $\bar{X} = \bar{k}e^{\lambda t}$ is the soln to the homogenous system: $\dot{\bar{X}} = A\bar{X}$, then

$$\dot{\bar{X}} = \bar{K}\lambda e^{\lambda t} = \bar{A}\bar{K}e^{\lambda t}$$

$$\Rightarrow \quad \bar{A}\bar{K} = \lambda\bar{K} \quad \text{or} \quad \bar{A}\bar{K} - \lambda\bar{K} = \bar{0} \quad (e^{\lambda t} \neq 0)$$

$$\Rightarrow \quad (\bar{A} - \lambda I)\bar{K} = \bar{0} \qquad \cdots \qquad (4.1a)$$

This eqn is equivalent to

$$\begin{aligned}
(a_{11} - \lambda)k_1 + a_{12}k_2 + \cdots + a_{1n}k_n &= 0 \\
a_{21}k_1 + (a_{22} - \lambda)k_2 + \cdots + a_{2n}k_n &= 0 \\
&\vdots \\
a_{n1}k_1 + a_{n2}k_2 + \cdots + (a_{nn} - \lambda)k_n &= 0 .
\end{aligned}$$

Avoiding the trivial soln $(k_1 = k_2 = \cdots = k_n = 0)$ we get

$$\det(\bar{A} - \lambda I) = 0 .$$

This is a polynomial in λ, which is called the characteristic eqn of the matrix $\bar{A}$. The solns $(\lambda_1, \lambda_2, \cdots, \lambda_n)$ are called the <u>eigenvalues</u> of $\bar{A}$.

A soln set $\bar{K} \neq \bar{0}$ of the corresponding values of λ is called an <u>eigenvector</u> of $\bar{A}$.

$\therefore$ we can now write the soln to the homogenous ODE as

$$\bar{X} = \bar{K}e^{\lambda t} .$$

We have three cases from the characteristic polynomials, which dependent on the values of λ:

Case 1: Distinct real eigenvalues.

When the values of λ are distinct $(\lambda_1, \lambda_2, \cdots, \lambda_n)$, then a set of linearly independent eigenvectors $\bar{k}_1, \bar{k}_2, \cdots, k_n$ can be found, and the soln be given by

$$\bar{x}_1 = \bar{k}_1 e^{\lambda_1 t}, \quad \bar{x}_2 = \bar{k}_2 e^{\lambda_2 t}, \cdots, \bar{x}_n = \bar{k}_n e^{\lambda_n t},$$

which is a fundamental set of solns on $(-\infty, \infty)$.

Case 2: Repeated eigenvalues:

If $m \in \mathbb{Z}^+$ and $(\lambda - \lambda_1)^m$ is a factor of the characteristic eqn, then λ_1 is said to be an eigenvalue of multiplicity m, that is the general is of the form

$$\bar{x}_1 = \bar{k}_{11} e^{\lambda_1 t}$$

$$\bar{x}_2 = \bar{k}_{21} t e^{\lambda_1 t} + \bar{k}_{22} e^{\lambda_1 t}$$

$$\vdots$$

$$\bar{x}_m = \bar{k}_{m1} \frac{t^{m-1}}{(m-1)!} e^{\lambda_1 t} + \bar{k}_{m2} \frac{t^{m-2}}{(m-2)!} + \cdots + \bar{k}_{mm} e^{\lambda_1 t}$$

where $\bar{K}_{ij}$ are vectors. This works fine provided λ_1 is the only root of characteristic polynomials. Otherwise, other cases should be incorporated.

Case 3: Complex eigenvalues.

Let $\lambda_1 = \alpha + i\beta$, where $\alpha, \beta \in \mathbb{R}$, be a complex eigenvalue of the coefficient matrix $\bar{A}$, and let $\bar{B}_1$ and $\bar{B}_2$ denote the column vectors. Then

$$\bar{X}_1 = [\bar{B}_1 \cos\beta t - \bar{B}_2 \sin\beta t] e^{\alpha t}$$

$$\bar{X}_2 = [\bar{B}_2 \cos\beta t + \bar{B}_1 \sin\beta t] e^{\alpha t}$$

are real linearly independent solns of $\dot{\bar{X}} = \bar{A}\bar{X}$.

Examples:

Find the general soln of the given system

1. $\dot{x} = 6x - y$
   $\dot{y} = 5x + 2y$

2. $\frac{dx}{dt} = x + y$
   $\frac{dy}{dt} = -2x - y$

3. $\frac{dx}{dt} = 5x + y$
   $\frac{dy}{dt} = -2x + 3y$

4. $\frac{dx}{dt} = 4x + 5y$
   $\frac{dy}{dt} = -2x + 6y$

S:
$$\dot{\bar{X}} = \begin{pmatrix} 1 & -1 & 2 \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{pmatrix} \bar{X}$$

Nonhomogeneous linear systems:

The general soln of a nonhomogeneous system $\dot{\bar{X}} = A\bar{X} + \bar{F}(t)$ on the interval I is given by $\bar{X} = \bar{X}_c + \bar{X}_p$, where $\bar{X}_c = c_1\bar{X}_1 + c_2\bar{X}_2 + \cdots + c_n\bar{X}_n$ is the complementary soln of $\dot{\bar{X}} = \bar{A}\bar{X}$, and $\bar{X}_p$ is any particular soln of the nonhomogeneous.

Methods of soln:

1) Undetermined coefficients:

As it was the case with previous ODEs, this method depends on educated guess of $\bar{F}(t)$.

The method of undetermined coefficients can be used to find a particular soln to the nonhomogeneous linear systems:

$$\bar{x}'(t) = \bar{A}\bar{x}(t) + \bar{f}(t),$$

where $\bar{A}$ is $n \times n$ constant matrix and

the entries of $\bar{f}(t)$ are polynomials, exponential fns, sines and cosines, or finite sums and products of these fns.

We proceed as we did for one DDE.

e.g.

Find the soln of

$$\bar{x}'(t) = A\bar{x}(t) + t\bar{g}, \text{ where}$$

$$A = \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \quad \text{and} \quad g = \begin{bmatrix} -9 \\ 0 \\ -18 \end{bmatrix}$$

Soln:

We start with complementary soln, i.e. soln to $\bar{x}'(t) = \bar{A}\bar{x}(t)$. This soln was previously found as

$$\bar{x}_c(t) = c_1 e^{-3t}\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} + c_2 e^{3t}\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} + c_3 e^{-3t}\begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}$$

Since $\bar{f}(t) = t\bar{g}$ is just a polynomial we are inclined to a particular soln of the form:

$$\bar{x}_p(t) = t\bar{a} + \bar{b}$$

$$\Rightarrow \bar{x}_p'(t) = \bar{a}$$

Substituting back into the system we get:

$$\bar{a} = \bar{A}(t\bar{a} + \bar{b}) + t\bar{g}$$

$$\Rightarrow \quad t(\bar{A}\bar{a} + \bar{g}) + (\bar{A}\bar{b} - \bar{a}) = \bar{0}$$

$$\Rightarrow \quad \begin{cases} \bar{A}\bar{a} = \bar{g} \\ \bar{A}\bar{b} = \bar{a} \end{cases}$$

Using the methods taught in second year in Linear algebra we are able to solve this system to get:

$$\bar{a} = \begin{bmatrix} 5 \\ 2 \\ 4 \end{bmatrix} \text{ and } \bar{b} = \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix}$$

$$\therefore \quad \bar{x}_p(t) = t\bar{a} + \bar{b} = t\begin{bmatrix} 5 \\ 2 \\ 4 \end{bmatrix} + \begin{bmatrix} 1 \\ 0 \\ 2 \end{bmatrix} = \begin{bmatrix} 5t+1 \\ 2t \\ 4t+2 \end{bmatrix}$$

Hence, the general ~~soln~~ is given by

$$\bar{x}(t) = \bar{x}_c(t) + \bar{x}_p(t)$$

More examples (some to be solved in the class):

Solve $\bar{x}'(t) = \bar{A}\bar{x}(t) + \bar{f}(t)$, where $\bar{A}$ and $\bar{f}(t)$ are given below:

(a) $\bar{A} = \begin{bmatrix} 6 & 1 \\ 4 & 3 \end{bmatrix}$, $\bar{f}(t) = \begin{bmatrix} -11 \\ -5 \end{bmatrix}$

(b) $\bar{A} = \begin{bmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, $\bar{f}(t) = e^t \begin{bmatrix} 2 \\ 4 \\ -2 \end{bmatrix}$

(c) $\bar{A} = \begin{bmatrix} 0 & -1 & 0 \\ -1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$, $\bar{f}(t) = \begin{bmatrix} e^{2t} \\ \sin(t) \\ t \end{bmatrix}$

Variation of parameters:

Let $\bar{x}_c(t)$ be the complementary soln for the homogeneous system:

$$\bar{x}'(t) = \bar{A}(t)\,\bar{x}(t),$$

where now the entries of $\bar{A}$ be any continuous fns of t. Because the general soln to the homogeneous case is given by $\bar{x}(t)\bar{c}$, we $\bar{c}$ is column matrix $(n \times 1)$, we seek a particular soln to the nonhomogeneous system:

$$\bar{x}'(t) = \bar{A}(t)\,\bar{x}(t) + \bar{f}(t) \quad \ldots (1)$$

of the form $\bar{x}_p'(t) = \bar{x}(t)\,\bar{v}(t)$

$$\Rightarrow \quad \bar{x}'(t) = \bar{x}(t)\,\bar{v}'(t) + \bar{x}'(t)\,\bar{v}(t)$$

Substituting into (1) above, we get

$$\bar{x}(t)\,\bar{v}(t) + \bar{x}'(t)\,v(t) = A(t)\,\bar{v}(t) + \bar{f}(t) \quad \ldots (2)$$

Since $\bar{x}(t)$ is the soln to the homogeneous case, the system in (2) reduces to

$$\bar{x}\,\bar{v}' + A\bar{x}\,\bar{v} = \bar{A}\bar{x}\bar{v} + \bar{f}$$

$$\Rightarrow \bar{x}\,\bar{v}' = \bar{f}$$

$$\Rightarrow \bar{v}(t) = \int \bar{x}^{-1}(t)\,\bar{f}(t)\,dt$$

$$\therefore \quad \bar{x}_p(t) = \bar{x}(t)\,\bar{v}(t) = \bar{x}(t)\int \bar{x}(t)^{-1}\bar{f}(t)\,dt$$

and $\bar{x}(t) = \bar{x}_c(t) + \bar{x}_p(t) = \bar{x}(t)\bar{c} + \bar{x}(t)\int \bar{x}^{-1}(t)\bar{f}(t)\,dt$

If the initial condition, $\bar{x}(t_0)=\bar{x}_0$ is given, then $\bar{c}=\bar{x}^{-1}(t_0)\bar{x}_0$

$$\Rightarrow \bar{x}(t)=\bar{x}(t)\bar{x}^{-1}(t_0)\bar{x}_0+\bar{x}(t)\int_{t_0}^{t}\bar{x}^{-1}(s)\bar{f}(s)\,ds.$$

Examples (some to be solved in the class):

Use the variation of parameters to find the general soln of $\bar{x}'(t)=\bar{A}\bar{x}(t)+\bar{f}(t)$ where

(a) $\bar{A}=\begin{bmatrix}0&1&1\\1&0&1\\1&1&0\end{bmatrix}$, $f(t)=\begin{bmatrix}3e^t\\-e^t\\-e^t\end{bmatrix}$

(b) $\bar{A}=\begin{bmatrix}1&-1&1\\0&0&1\\0&-1&2\end{bmatrix}$, $\bar{f}(t)=\begin{bmatrix}0\\e^t\\e^t\end{bmatrix}$

(c) $\bar{A}=\begin{bmatrix}0&1&0&0\\-1&0&0&0\\0&0&0&1\\0&0&1&0\end{bmatrix}$, $\bar{f}(t)=\begin{bmatrix}t\\0\\e^t\\t\end{bmatrix}$

Sometimes a second order ODE or higher order ODE can be converted into a system of first order ODEs and use the methods discussed above to solve it.

Starting with $x^{(n)}=f(t,x,x',x'',\cdots,x^{(n-1)})$,

let $x_1=x \Rightarrow x_1'=x'=x_2$

$$x_2' = x'' = x_3$$

$$\vdots$$

$$x_{n-1}' = x^{(n-1)} = x_n$$

$$x_n' = x^{(n)} = f(t, x_1, x_2, \cdots, x_{n-1})$$

e.g.

Convert $y''' - 6y'' + 2y' + y = 0$ to a system of ODEs.

Soln: $y''' = 6y'' - 2y' - y$

let $x_1 = y$,

$$x_1' = y' = x_2, \quad x_2' = y'' = x_3$$

$$x_3' = y''' = 6x_3 - 2x_2 - x_1$$

∴ Our system is:

$$\boxed{\begin{aligned} x_1' &= x_2 \\ x_2' &= x_3 \\ x_3' &= 6x_3 - 2x_2 - x_1 \end{aligned}}$$

which can be written in a matrix form as

$$\bar{x}' = \begin{bmatrix} x_1' \\ x_2' \\ x_3' \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & -2 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

We end this section by giving some of the explanations on the repeated cases we met on Eigenvalue problem section.

Given a homogeneous system

$\vec{x}'(t) = \bar{A}\vec{x}(t)$, where $\bar{A}$ is a constant $n \times n$ matrix, the soln is given by $\vec{x}(t) = e^{\bar{A}t}\vec{c}$.

Now consider $e^{\bar{A}t}$:

$$e^{\bar{A}t} = I + \bar{A}t + \frac{\bar{A}^2 t^2}{2!} + \cdots + \frac{\bar{A}^n t^n}{n!} + \cdots$$

(remember: $e^{ax} = 1 + ax + \frac{(ax)^2}{2!} + \frac{(ax)^3}{3!} + \cdots + \frac{(ax)^n}{n!} + \cdots$)

If $\bar{A}$ is a diagonal matrix, then the computation $e^{\bar{A}t}$ is easier, e.g.

If $\bar{A} = \begin{bmatrix} -1 & 0 \\ 0 & 2 \end{bmatrix}$, then $\bar{A}^2 = \bar{A}\bar{A} = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$

and $\bar{A}^3 = \bar{A}^2\bar{A} = \begin{bmatrix} -1 & 0 \\ 0 & 8 \end{bmatrix}$, $\cdots$,

$$\bar{A}^n = \begin{bmatrix} (-1)^n & 0 \\ 0 & 2^n \end{bmatrix}$$

and so

$$e^{\bar{A}t} = \sum_{n=0}^{\infty} \frac{\bar{A}^n t^n}{n!} = \begin{bmatrix} \sum_{n=0}^{\infty} \frac{(-1)^n t^n}{n!} & 0 \\ 0 & \sum_{n=0}^{\infty} \frac{2^n t^n}{n!} \end{bmatrix}$$

$$= \begin{bmatrix} e^{-t} & 0 \\ 0 & e^{2t} \end{bmatrix}$$

$$\frac{d}{dt}\left(e^{\bar{A}t}\right) = \frac{d}{dt}\left[I + \bar{A}t + \frac{\bar{A}^2 t^2}{2!} + \cdots + \frac{\bar{A}^n t^n}{n!} + \cdots\right]$$

$$= \bar{A} + \bar{A}^2 t + \frac{\bar{A}^3 t^2}{2} + \cdots + \frac{\bar{A}^n t^{n-1}}{(n-1)!} + \cdots$$

$$= A\left[I + At + \frac{A^2t^2}{2!} + \cdots + \frac{A^{n-1}t^{n-1}}{(n-1)!} + \cdots\right]$$

Hence,

$$\frac{d}{dt}\left(e^{\bar{A}t}\right) = \bar{A}\,e^{\bar{A}t}$$

& so e^{At} is the soln to $\bar{x}' = \bar{A}\bar{x}$